

# Construction of Longer-Horizon Variables

## 1. Futures Returns (FutRet)

The original four-week cumulative futures return,  $\text{FutRet}_{t4}$ , is computed multiplicatively from weekly returns:

$$\text{FutRet}_{t4} = (1 + r_t)(1 + r_{t-1})(1 + r_{t-2})(1 + r_{t-3}) - 1,$$

where  $r_t$  is the one-week return of the front-month contract at week  $t$ . Expressed in percent, the result is multiplied by 100.

### 24-Week Horizon

$$\text{FutRet}_{t24} = \frac{\text{FutRet}_{t8} \times L^8 \text{FutRet}_{t8} \times L^{16} \text{FutRet}_{t8}}{100^2},$$

compounding three disjoint eight-week returns to cover 24 weeks.

### 52-Week Horizon

$$\text{FutRet}_{t52} = \frac{\prod_{j=0}^{12} L^{4j} \text{FutRet}_{t4}}{100^{12}},$$

since 52 weeks is covered by thirteen successive 4-week slices.

## 2. Spot and Equity Returns, and Volatility Changes

Let  $X \in \{\text{DSpot}, \text{xomRet}, \text{bpRet}, \text{rdsaRet}, \text{DOilVol}\}$  be the variables with original four-week values  $X_{t4}$ . Each longer horizon is constructed by *adding* disjoint four-week changes:

### 24-Week Horizon

$$X_{t24} = X_{t8} + L^8 X_{t8} + L^{16} X_{t8},$$

where  $X_{t8} = X_{t4} + L^4 X_{t4}$ .

### 52-Week Horizon

$$X_{t52} = \sum_{j=0}^{12} L^{4j} X_{t4}.$$

## 3. World Industrial Production Index (WIPI)

Monthly WIPI growth over eight weeks,  $\text{WIPI\_8wk\_monthly}$ , yields:

$$\text{WIPI}_{24wk} = L^{16} \text{WIPI\_8wk\_monthly}, \quad \text{WIPI}_{52wk} = L^{44} \text{WIPI\_8wk\_monthly}.$$

## 4. Physical Variables

### 4.1. Oil Production (DProd)

Define, for any integer  $n \geq 1$ ,

$$DProd_{t,4n} = 100 \ln \left( \frac{\sum_{i=0}^{n-1} S_i(t)}{n S_n(t)} \right), \quad S_i(t) = \sum_{k=0}^3 \text{prod}_{t-4i-k}, \quad (1)$$

so each  $S_i(t)$  is the total production in the  $i$ -th past 4-week block (counting backward from calendar date  $t$ ).

For a single block ( $n = 1$ ) this gives

$$DProd_{t,4} = 100 \ln \left( \frac{S_0(t)}{S_1(t)} \right), \quad (2)$$

a log-ratio of the current 4-week total to the preceding one.

Define the building-block increment

$$r_i(t) \equiv \ln \frac{S_i(t)}{S_{i+1}(t)} = \frac{1}{100} DProd_{t-4i,4} = \frac{1}{100} L^{4i} DProd_{t,4}, \quad (3)$$

where  $L^{4i}$  denotes a  $4i$ -week lag.

**1. Rewrite the Sum:** Because

$$S_i = S_n \exp \left( \sum_{k=i}^{n-1} r_k \right),$$

we have

$$\sum_{i=0}^{n-1} S_i = S_n \sum_{i=0}^{n-1} \exp \left( \sum_{k=i}^{n-1} r_k \right) = S_n \sum_{i=0}^{n-1} e^{c_i}, \quad c_i = \sum_{k=i}^{n-1} r_k.$$

**2. Plug into the Target:**

$$DProd_{t,4n} = 100 \left[ \ln S_n + \ln \left( \sum_{i=0}^{n-1} e^{c_i} \right) - \ln n - \ln S_n \right] = 100 \left[ \ln \left( \sum_{i=0}^{n-1} e^{c_i} \right) - \ln n \right].$$

This exactly matches the Stata loops for  $n = 6$  (24 weeks) and  $n = 13$  (52 weeks). For  $n = 2$ , it reduces to

$$DProd_{t8} = 100 \left[ \ln(e^{r_0} + 1) + r_1 - \ln 2 \right],$$

recovering the eight-week formula.

### 4.2. Oil Inventories (DInv)

$$DInv_{t24} = DInv_{t8} + L^8 DInv_{t8} + L^{16} DInv_{t8}, \quad DInv_{t52} = \sum_{j=0}^{12} L^{4j} DInv_{t4}.$$

**Rationale:** Longer horizons capture medium- and long-term trends by aggregating short-term dynamics. Multiplicative compounding is used for returns, additive for log-differences and linear changes, ensuring consistency with underlying economic processes.